

ANALOGUE CIRCUIT SIMULATION

Circuit simulators: A practical approach to electronic circuit designs

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Simulators play a vital role in electronics. Before constructing any real circuit, different components need to be connected to check outputs for their corresponding inputs. It is impractical to construct the entire circuit in real-time and then test it. The reason: every time a circuit is tested, it is being done under different conditions, posing a risk of damage to the components. Besides, trying out different circuit ideas on a breadboard is also impractical and time-consuming. Simulators solve these problems effectively. SPICE is one such analogue circuit simulator that supports many other software.

But, before selecting an analogue circuit simulator, one should look for features like: What are the different components supported by the software? Does it support other component models from different software? How good is the waveform viewer? Does the software give complete simulation control? What is the cost of the software? Whether the software is useful for professional or educational purposes? Let us consider some of the commonly-used analogue circuit simulation software.

Aim-Spice

Aim-Spice, aka Automatic Integrated Circuit Modeling Spice, is a clone of the original SPICE simulator, which was developed in the University of California, Berkeley. It contains models for most of the circuit elements and can handle non-linear circuits too. The latest version of the software available is 5.5.

What's Special?

Aim-Spice supports features like BSIM4v6.5 and is now compatible with Windows 7. Besides, it also supports Linux platform. The software uses Berkeley SPICE version 3.E1 as the kernel and has new models for the latest semiconductor devices. It also has a graphical postprocessor and AIM-postprocessor. The advanced models that have been implemented in this software are Heterostructure Diode, Heterojunction Bipolar Transistor (HBT) MOSFETs.

The extensions of Berkeley SPICE model in Aim-Spice are temperature sweep analysis, support for libraries, enhanced commenting features and support for extended commands, such as .connect and .plot and others.

Users' feedback

The users have complete control while simulating any circuit because they have to first select the parameters for simulation. Besides, the output of the simulated circuit can be viewed in graphical mode. The software also has a wide collection of device models.

Areas of improvement

The charge storage in MOS devices, based on the Meyer model, is incorrectly calculated. Also, the base node of the bipolar transistor (BJT) is incorrectly modeled.

Home page: <http://www.aimspice.com>
Download URL: <http://www.aimspice.com/download.html>
Platform supported: Windows and Linux

CIRSIM

CIRSIM is a software for simulating programs. The current version of CIRSIM is 3.08.

What's special?

CIRSIM is used to calculate the steady state performance of networks with up to 100 nodes. The software can be used for any combination of active and passive components, including transformers and theoretical values. Its output can be displayed in graphical format. CIRSIM can be used to check the performance of non-linear audio, video, RF amplifier, oscillator, etc, circuits. The new version has better help facilities and improvements in text editor using RTF for input of circuit details.

According to the author, the competitor of CIRSIM is ACIRAN.

Users' feedback

The selling point of this software is its cost, which is much lesser than of any commercially available linear circuit analysis software. Its results too are quite accurate. Another attractive feature is its graph plot. Users may plot various graphs using one set of component values or nodes. They may change these values and plot a superimposed plot in a different colour.

Areas of improvement

The demo version of the software is fully functional but limited to six nodes. Some more nodes could help the users to test the software better.

Home page: <http://www.bells-hill.freesevice.co.uk/>
Download URL: <http://www.bells-hill.freesevice.co.uk/download.htm>
Platform supported: DOS and Windows

MicroCAP

MicroCAP has been developed by Spectrum Software; the latest version available is 9.0.7.0.